HIMANSHU SINGH YADAV

AI & ML ENGINEER

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PROFESSIONAL SUMMARY

AI & ML Engineer with hands-on expertise building production-grade Machine Learning pipelines, RESTful APIs, and responsive web applications. Demonstrated track record in architecting NLP, recommendation engines, and modern frontend interfaces using Python, React 19, Next.js 16, and Flask. Specializing in bridging the gap between raw data models and user-facing cloud software. Passionate about turning complex algorithms into intuitive, high-value user products.

WORK EXPERIENCE

Machine Learning Intern | Skillbit Technologies — *Oct 2025 – Nov 2025*

- Executed end-to-end ML pipelines on real-world industry datasets: data collection, preprocessing, feature engineering, model training, and evaluation.
- Built a Movie Recommendation Engine using item-based collaborative filtering and cosine similarity on the MovieLens dataset and Developed an SMS Spam Classifier (Naive Bayes + TF-IDF) on the UCI SMS Spam Collection.

PROJECTS

CorAI – Heart Disease Prediction System | [Github]

- Built a full-stack web app with Flask & SQLite—REST-style routes, a doctor login system, and a patient CRUD dashboard—serving an ML model that classifies heart disease risk from 11 medical parameters.
- Integrated the Google Gemini API for an in-app chatbot and automated PDF report generation via ReportLab.

Vaelos – Smart Transport Operations Platform | [Github]

- Built a full-stack PWA for fleet management featuring real-time vehicle mapping, automated trip dispatching, and driver compliance tracking.
- Implemented a rule-based AI predictive maintenance algorithm scoring asset risk using telemetry data (odometer, fuel drift, service history) and Integrated live WebSocket metrics updates, plain-English voice/text AI assistant queries, and comprehensive audit logging.

MindPulse – AI-Powered Mental Health Analyzer | [Github]

- Developed a Flask app classifying journal entries into 6 emotion categories with a TF-IDF + scikit-learn pipeline, achieving 82% validation accuracy, with Matplotlib-based mood-trend visualizations.
- Applied an NLTK preprocessing pipeline (tokenization, lemmatization, stop-word removal) and visualized per-session mood trends over weeks to help users track emotional well-being.

TECHNICAL SKILLS

AI & ML: Deep Learning, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, NLTK, NLP
Backend & APIs: Flask, REST APIs, SQLite, Node.js, Express.js, WebSockets, JWT Auth
Frontend: JavaScript, TypeScript, React 19, Next.js 16, Tailwind CSS, HTML5, Modern CSS
Cloud & Tools: AWS Cloud, Vercel, Railway, Render, Git & GitHub, VS Code, Power BI, Streamlit
Core Fundamentals: Object-Oriented Programming (OOP), Data Preprocessing, Feature Engineering, Model Evaluation, API Integration

EDUCATION

B.Tech – Artificial Intelligence & Machine Learning — *Expected May 2027*

P. P. Savani University, Kosamba, Gujarat — **CGPA: 7.71 / 10.00**

ACHIEVEMENT

- **SSIP (2025)** — Selected under Student Startup & Innovation Policy at P P Savani University for developing an **AI-powered Crop Yield Prediction System** using ML on soil, weather & fertilizer data to support data-driven agricultural decisions for farmers.